

Retail Analysis Project

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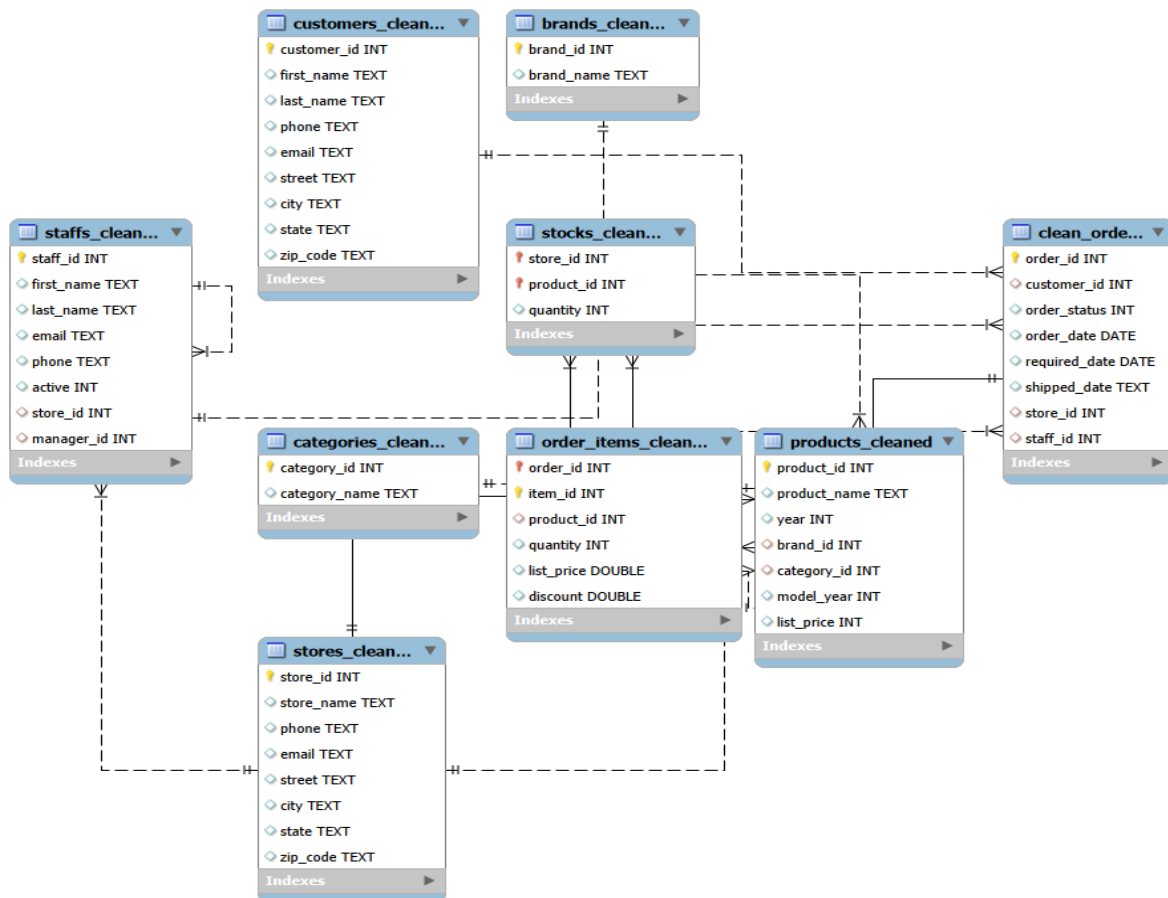
Tools Used:

Google Colab.

MYSQL.

Power BI.

Entity Relationship(ER) Diagram:



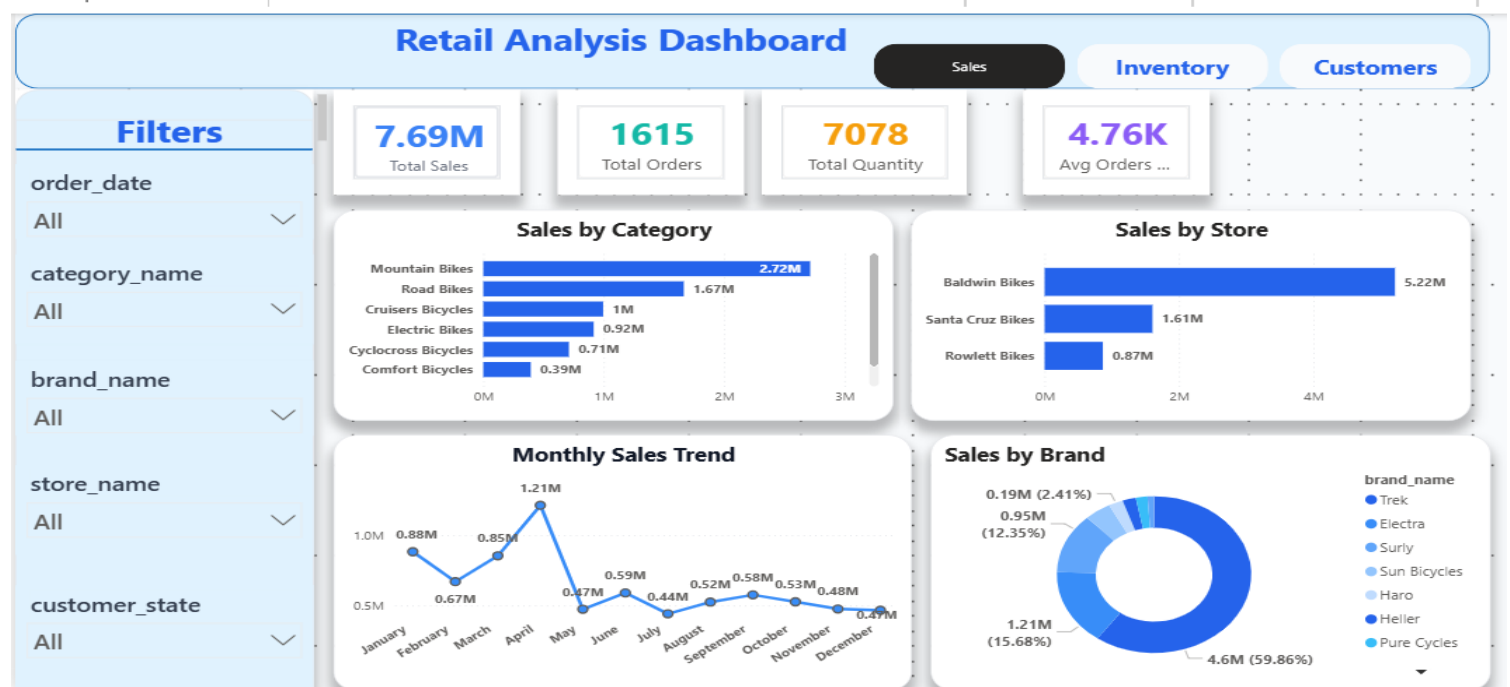
- Customers – customer details.
- Orders – order information.
- Order Items – product-level sales.
- Products-product details.
- Brands – brand details.
- Categories – product categories.
- Stocks – inventory quantity.

- Stores – store information.
- Staffs- employee details.

Important Relationships:

- One customer can place many orders.
- One order can contain many products.
- One product belongs to one category and one brand.
- One store can have many staff members.
- Stocks table connects stores and products.

Sales Dashboard Summary:



Insights:

- Mountain Bikes generated the highest sales.
- Road Bikes and Cruisers Bicycles also contributed significantly.

- Comfort and Children Bicycles recorded the lowest among categories.
- Baldwin Bikes is top-performing state.
- Rowlett Bikes generated the lowest sales.

Monthly Sales Trend:

- Sales peaked during April.
- Sales fluctuated across months after April.
- Lower sales were observed toward the end of the year.

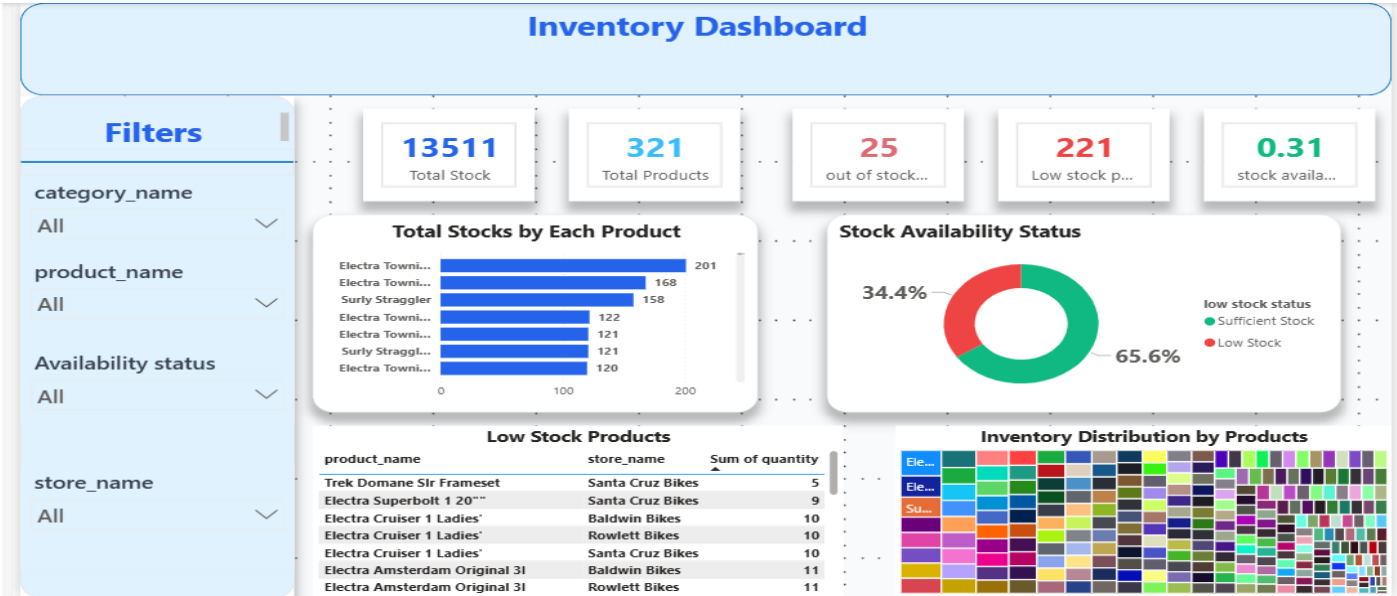
Sales by Brand:

- Trek brand contributes the highest share of total sales.
- Electra and Surly brands also showed good performance.
- Other brands contributed smaller portions of revenue.

Conclusions:

- The dashboard clearly highlights top-performing stores, brands and product categories.
- It helps management monitor sales trends, identify business opportunities, and improve strategic decision-making.

Inventory Dashboard Summary:



Insights:

Total Stocks by Each Product:

- Some products maintain very high stock quantities.
- Electra and Surly products appear frequently among high-stock items.
- Stock distribution varies across products.

Stock Availability Status:

- Majority of products are sufficiently stocked.
- Around 34.4% of products fall under low-stock condition.
- Low-stock products require immediate monitoring.

Low-stock Products:

- Several products have critically low inventory levels.
- Some stores have stock quantities below minimum levels.
- Products with low stock may affect the future sales performance for the business.

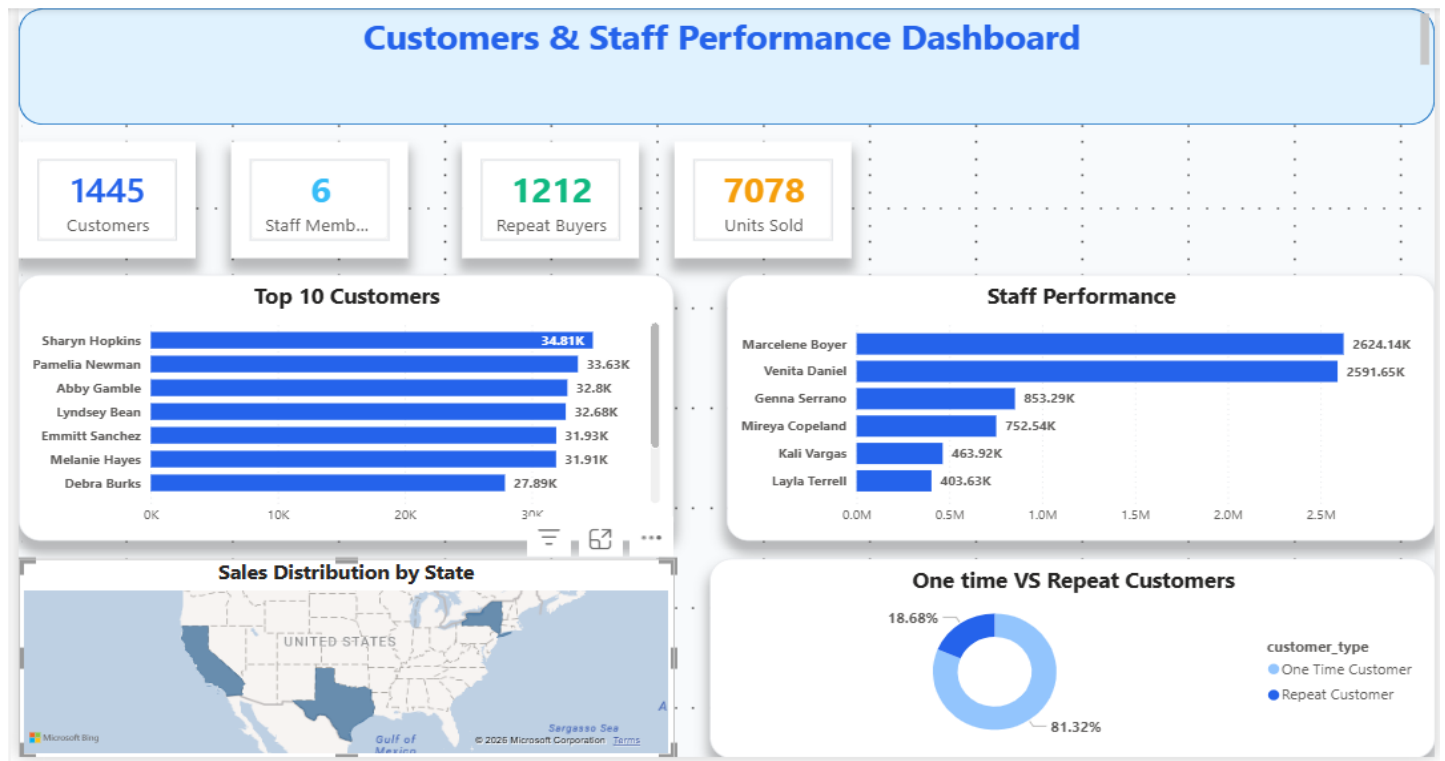
Inventory Distribution by Products:

- Inventory is distributed across multiple products and brands.
- Certain products occupy a larger share of inventory.
- Uneven stock distribution can lead to overstock or shortages.

Conclusions:

- This dashboard helps track stock levels, identify low-stock products, and monitor inventory distribution across stores.
- It enables better inventory planning timely restocking, and efficient stock management.
- To avoid product shortages and improve business operations.

Customers & Staff Performance Dashboard Summary:



Insights:

Top 10 Customers:

- The top customers generated significant revenue for the business.
- Sharyn Hopkins is the top customer in spending.
- The top customer group contributes a major share of sales revenue.

Staff Performance:

- Marcelene Boyer achieved highest sales performance.
- Venita Daniel also contributed strong sales results.
- Staff sales performance varies significantly across employees.

One-time vs Repeat Customers:

- Around 81.32% customers are one-time Customers.
- Only 18.68% customers are Repeat Customers.
- Customer retention appears lower compared to the one-time purchases.
- The business has opportunities to improve customer loyalty.

Geographical Sales Analysis:

- New York (NY) generated the highest sales revenue.
- The State contributed the largest share of total business revenue.
- Other two states recorded comparatively lower sales performance.

Conclusions:

- This Dashboard analyzes customer behavior, staff contribution, and geographical sales performance.
- The geographical sales analysis showed that New York generated the highest sales revenue of 52,15,790.42 making it the strongest business region.

Business Recommendations:

- Increase marketing and promotions for high-selling categories like Mountain Bikes.
- Maintain sufficient inventory for top-performing brands such as Trek.
- Apply successful sales strategies from high-performing stores to low-performing strategies.
- Restock the products immediately to avoid sales loss.
- Reduce excess inventory for slow-moving products to minimize the costs.
- Introduce loyalty programs to increase repeat customers.
- Use personalized marketing campaigns for high value customers.
- Allocate inventory based on store demand and sales performance.
- Improve customer engagement through better after sales service and feedback collection.
- Reward top-performing staff members with incentives and recognition programs.
- Provide some training and support for low performing employees to improve sales performance.
- Focus on improving customer satisfaction, operational efficiency and long-term business growth.